mergencystretch

Chapter 5. Partial Identification and Soft Interventions

Natural bounds, the instrumental-variable linear program, stochastic interventions and sigma-calculus

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5.1 Partial identification

Motivation. The central question is what can still be said about $P(y \mid \mathrm{do}(x))$ when it is not identifiable: instead of a point, we seek the tightest interval $[a(y;x), b(y;x)] \subseteq [0,1]$ consistent with the data and the graph.

5.1.1 The partial identification problem

Motivation. Extend the yes/no identification question to bounds.

Recall that $P(y \mid \mathrm{do}(x))$ is identifiable if it is uniquely determined by $P(\mathbf{v})$ and $\mathcal{G}$. When it is not, an inference engine given the query, the graph and data such as $P(x,y,z)$ answers "no". We then ask whether a non-trivial bound can be given:

$$P(y \mid \mathrm{do}(x)) \in [a(y;x),\ b(y;x)] \subseteq [0,1].$$

Definition (Partial identifiability)

The effect of X on Y is partially identifiable given $\mathcal{G}$ if $P(\mathbf{V})$ determines a bound $P(y \mid \mathrm{do}(x)) \in [a,b]$ that is a proper subset of $[0,1]$.

Identifiable $\Rightarrow$ partially identifiable (with $a = b$, ignoring the degenerate case), but not conversely.

5.1.2 Natural bounds

Motivation. Derive the simplest bounds, valid for any confounded pair, and show they cannot be improved without more structure.

Theorem 5.1 (Natural bounds; Manski 1990, Robins 1989)

For an SCM compatible with $X \to Y$, $X \leftrightarrow Y$ (latent U), given $P(x,y)$,

$$a(y;x) = P(y,x) \ \le\ P(y \mid \mathrm{do}(x)) \ \le\ P(y,x) + 1 - P(x) = b(y;x).$$

Proof [Supplement]. With $X \leftarrow f_X(u)$, $Y \leftarrow f_Y(x,u)$, the truncated product gives

$$\begin{aligned} P(y \mid \mathrm{do}(x)) &= \sum_u P(y \mid x,u)P(u) = \sum_u P(y \mid x,u)\big(P(u,x) + P(u) - P(u,x)\big) \\ &= \sum_u P(y,x,u) + \sum_u P(y \mid x,u)\big(P(u) - P(u,x)\big) \\ &= P(y,x) + \sum_u P(y \mid x,u)\big(P(u) - P(u,x)\big). \end{aligned}$$

Since $0 \le P(y \mid x,u) \le 1$ and $P(u) - P(u,x) \ge 0$, the last sum lies between 0 and $\sum_u (P(u) - P(u,x)) = 1 - P(x)$. ∎

The width of the interval is $1 - P(x)$: the probability mass of units that did not receive x, about whom the data say nothing under $\mathrm{do}(x)$. Equivalently $b(y;x) = 1 - P(x,y')$ with y' the complement of y.

Example 5.1

$X, Y, U \in \{0,1\}$, $X \leftarrow U$, $Y \leftarrow X \oplus U$, $P(U=0) = 0.9$. Observationally $X = U$ and $Y = 0$ always: $P(X=0, Y=0) = 0.9$, $P(X=1, Y=0) = 0.1$. Under $\mathrm{do}(X=x)$, $Y = x \oplus U$:

	$P(Y=1 \mid \mathrm{do}(x))$	$a(Y=1;x)$	$b(Y=1;x)$
$x=0$	0.1	0	0.1
$x=1$	0.9	0	0.9

The true effects lie on the upper bounds.

5.1.2.1 Tightness

Theorem 5.2

For binary X, Y and any $P(x,y)$ there exist SCMs $\mathcal{M}^{(1)}, \mathcal{M}^{(2)}$ with $P^{(1)}(x,y) = P^{(2)}(x,y) = P(x,y)$ and $P^{(1)}(y \mid \mathrm{do}(x)) = a(y;x)$, $P^{(2)}(y \mid \mathrm{do}(x)) = b(y;x)$.

Proof [Supplement]. Take any compatible $\mathcal{M}$ ($X \leftarrow f_X(u)$, $Y \leftarrow f_Y(x,u)$) inducing P, and define, with the same $P(U)$,

$$\mathcal{F}^{(1)}: \; Y \leftarrow \begin{cases} f_Y(x,u) & x = f_X(u) \\ 0 & \text{otherwise} \end{cases}, \qquad \mathcal{F}^{(2)}: \; Y \leftarrow \begin{cases} f_Y(x,u) & x = f_X(u) \\ 1 & \text{otherwise} \end{cases},$$

both with $X \leftarrow f_X(u)$. Without intervention $X = f_X(U)$ always, so $Y = f_Y(X,U)$ in both models and $P^{(i)}(x,y) = \sum_u P(x \mid u) P(y \mid x,u) P(u) = P(y,x)$. Under $\mathrm{do}(X=x)$, U and x need not match:

$$\begin{aligned} P^{(i)}(Y=1 \mid \mathrm{do}(x)) &= P(Y=1 \mid x, f_X(U)=x) P(f_X(U)=x) \\ &\quad + P^{(i)}(Y=1 \mid x, f_X(U) \neq x) P(f_X(U) \neq x) \\ &= P(x, Y=1) + \mathbb{1}[i=2](1 - P(x)). \end{aligned}$$

Thus $\mathcal{M}^{(1)}$ attains a and $\mathcal{M}^{(2)}$ attains b (for $Y=1$; $Y=0$ is symmetric). ■

5.1.2.2 Limitations

Natural bounds are tight for the bow graph with binary variables, but not when further structure is available, e.g. an instrument Z. Can the graph be exploited for tighter bounds? Yes: canonical type models and linear programming.

5.1.3 Canonical type models and LP bounds

Motivation. Replace the unknown functional forms by a finite set of response types, turning the bounding problem into a linear program.

5.1.3.1 Experiments with non-compliance

Setting: treatment assigned Z (randomized, the instrument), treatment realized X, primary outcome Y, and exogenous U confounding X and Y: $Z \to X \to Y$, $X \leftrightarrow Y$. Task: given $P(x,y \mid z)$, bound $P(y \mid \mathrm{do}(x))$.

Theorem 5.3 (Natural bounds in the IV model)

$$a(y;x) = \max_z P(y,x \mid z), \qquad b(y;x) = \min_z \left[P(y,x \mid z) + 1 - P(x \mid z)\right].$$

Proof [Supplement]. Fix z. In the submodel $\mathcal{M}_z$, the natural bounds give $P(y,x \mid \text{do}(z)) \leq P(y \mid \text{do}(x,z)) \leq P(y,x \mid \text{do}(z)) + 1 - P(x \mid \text{do}(z))$. By exclusion ($Z$ affects Y only through X), $P(y \mid \text{do}(x,z)) = P(y \mid \text{do}(x))$; by randomization, $P(\cdot \mid \text{do}(z)) = P(\cdot \mid z)$. So $P(y,x \mid z) \leq P(y \mid \text{do}(x)) \leq P(y,x \mid z) + 1 - P(x \mid z)$ for every z, and taking the maximum of the lower and the minimum of the upper bounds over z gives the result. ■

They are not tight in IV models.

5.1.3.2 Optimization perspective

Let $\mathcal{M}$ be the set of SCMs compatible with the IV graph: $Z \leftarrow f_Z(U_Z)$, $X \leftarrow f_X(Z,U)$, $Y \leftarrow f_Y(X,U)$. The sharp bounds are

$$a(y;x) = \min_{M \in \mathcal{M}:\, P_M(x,y|z) = P(x,y|z)} P_M(y \mid \text{do}(x)), \qquad b(y;x) = \max_{M \in \mathcal{M}:\, P_M(x,y|z) = P(x,y|z)} P_M(y \mid \text{do}(x)).$$

This is hard to solve directly because the forms of $\mathcal{F}$ and $P(\mathbf{U})$ are unspecified.

5.1.3.3 U as a selector and response variables

For $X \leftarrow f_X(Z,U)$ and fixed $U = u$, $f_X(\cdot,u)$ is a deterministic function from Z to X: U *selects* which mapping $Z \to X$ is in effect. The domain $\mathfrak{X}_U$ can therefore be partitioned by the induced function. With binary Z, X there are $|\mathfrak{X}_X|^{|\mathfrak{X}_Z|} = 2^2 = 4$ functions:

$$\{u : f_X(0,u) = 0,\ f_X(1,u) = 0\} \quad (X = 0), \qquad \{u : f_X(0,u) = 0,\ f_X(1,u) = 1\} \quad (X = Z),$$
$$\{u : f_X(0,u) = 1,\ f_X(1,u) = 0\} \quad (X = 1 - Z), \qquad \{u : f_X(0,u) = 1,\ f_X(1,u) = 1\} \quad (X = 1).$$

(With ternary Z and binary X there would be 2^3.) A *response variable* indexes this partition:

$R_X = 0$: $X = 0$ (never-taker) $R_X = 1$: $X = Z$ (complier) $R_X = 2$: $X = 1 - Z$ (defier)
$R_X = 3$: $X = 1$ (always-taker)

Similarly $R_Y \in \{0,1,2,3\}$ for $Y \leftarrow f_Y(X,U)$: never-recover ($Y = 0$), helped ($Y = X$), hurt ($Y = 1 - X$), always-recover ($Y = 1$). Together (R_X, R_Y) partition $\mathfrak{X}_U$ into $4 \times 4 = 16$ cells with probabilities

$$q_{j,k} = P(R_X = j, R_Y = k), \qquad \sum_{j=0}^{3} \sum_{k=0}^{3} q_{j,k} = 1.$$

Definition (Canonical type model)

An SCM with $X, Y, Z \in \{0,1\}$ and $\mathbf{U} = \{R_X, R_Y, R_Z\}$, $R_X, R_Y \in \{0,1,2,3\}$, $R_Z \in \{0,1\}$:

$$Z \leftarrow r_Z, \qquad X \leftarrow f_X(z, r_X) = \begin{cases} 0 & r_X = 0 \\ z & r_X = 1 \\ 1 - z & r_X = 2 \\ 1 & r_X = 3 \end{cases}, \qquad Y \leftarrow f_Y(x, r_Y) = \begin{cases} 0 & r_Y = 0 \\ x & r_Y = 1 \\ 1 - x & r_Y = 2 \\ 1 & r_Y = 3 \end{cases}.$$

Theorem 5.4 (Correspondence; Balke and Pearl 1997)

For any IV model $\mathcal{M}_1$ there is a canonical type model $\mathcal{M}_2$ with $P^{\mathcal{M}_1}(x, y \mid z) = P^{\mathcal{M}_2}(x, y \mid z)$ and $P^{\mathcal{M}_1}(y \mid \mathrm{do}(x)) = P^{\mathcal{M}_2}(y \mid \mathrm{do}(x))$.

Proof [Supplement]. Partition the domain of U by the pair of response functions it induces: $u \mapsto (x(\cdot), y(\cdot))$ with $x(z) = f_X(z, u)$ and $y(x) = f_Y(x, u)$. With binary variables there are 4 functions $z \mapsto x$ and 4 functions $x \mapsto y$, so 16 cells. Define $\mathcal{M}_2$ with a discrete $R = (R_X, R_Y)$ taking value (j, k) with probability $q_{j,k}$, the $P(U)$-mass of the corresponding cell, and $X = \mathrm{response}_j(Z)$, $Y = \mathrm{response}_k(X)$. For each z, $P^{\mathcal{M}_1}(x, y \mid z)$ is the mass of the u's with $f_X(z, u) = x$ and $f_Y(x, u) = y$, which is a union of cells, so it equals $P^{\mathcal{M}_2}(x, y \mid z)$; likewise $P^{\mathcal{M}_1}(y \mid \mathrm{do}(x))$ is the mass of $\{u : f_Y(x, u) = y\}$, a union of cells, and equals $P^{\mathcal{M}_2}(y \mid \mathrm{do}(x))$. ■

(Set $q_{j,k}$ to the $P(U)$-mass of the cell of $\mathfrak{X}_U$ inducing the pair of functions (j, k).) Hence, with $\mathcal{M}$ now the canonical models,

$$a(y; x) = \min_{M \in \mathcal{M}} P_M(y \mid \mathrm{do}(x)), \quad b(y; x) = \max_{M \in \mathcal{M}} P_M(y \mid \mathrm{do}(x)) \quad \text{s.t. } P_M(x, y \mid z) = P(x, y \mid z).$$

Summary. U is partitioned into response types by the functions it induces; for the binary IV, R_X (never-taker, complier, defier, always-taker) and R_Y; the joint $P(R_X, R_Y)$ has 16 parameters $q_{j,k}$; every IV model has an equivalent canonical model. Bounding $P(y \mid \mathrm{do}(x))$ reduces to a linear program over $\mathbf{q} = (q_{j,k}) \in \mathbb{R}^{16}$.

5.1.3.4 LP formulation

Observational constraints. Let $p_{yx.z} = P(Y = y, X = x \mid Z = z)$. Each is a linear function of $\mathbf{q}$, e.g.

$$\begin{aligned} p_{10.1} = P(Y = 1, X = 0 \mid Z = 1) &= P(\text{never-taker, hurt}) + P(\text{never-taker, always-recover}) \\ &\quad + P(\text{defier, hurt}) + P(\text{defier, always-recover}) \\ &= q_{0,2} + q_{0,3} + q_{2,2} + q_{2,3}. \end{aligned}$$

(With $Z = 1$, $X = 0$ requires $R_X \in \{0, 2\}$; then $Y = 1$ at $X = 0$ requires $R_Y \in \{2, 3\}$.) In full:

$$\begin{aligned} p_{00.0} &= q_{0,0} + q_{0,1} + q_{1,0} + q_{1,1}, & p_{00.1} &= q_{0,0} + q_{0,1} + q_{2,0} + q_{2,1}, \\ p_{01.0} &= q_{2,0} + q_{2,2} + q_{3,0} + q_{3,2}, & p_{01.1} &= q_{1,0} + q_{1,2} + q_{3,0} + q_{3,2}, \\ p_{10.0} &= q_{0,2} + q_{0,3} + q_{1,2} + q_{1,3}, & p_{10.1} &= q_{0,2} + q_{0,3} + q_{2,2} + q_{2,3}, \\ p_{11.0} &= q_{2,1} + q_{2,3} + q_{3,1} + q_{3,3}, & p_{11.1} &= q_{1,1} + q_{1,3} + q_{3,1} + q_{3,3}. \end{aligned}$$

Causal effect. Y under $\mathrm{do}(X = 0)$ equals 1 iff $R_Y \in \{\text{hurt, always-recover}\}$:

$$P(Y = 1 \mid \mathrm{do}(X = 0)) = P(R_Y = 2) + P(R_Y = 3) = \textstyle\sum_j (q_{j,2} + q_{j,3}),$$

$$P(Y = 1 \mid \mathrm{do}(X = 1)) = \textstyle\sum_j (q_{j,1} + q_{j,3}).$$

Optimization. The bound $P(Y = 1 \mid \mathrm{do}(X = 0)) \in [a_{10}, b_{10}]$ solves

$$a_{10} = \min_{\mathbf{q}} \sum_{j=0}^{3} (q_{j,2} + q_{j,3}), \qquad b_{10} = \max_{\mathbf{q}} \sum_{j=0}^{3} (q_{j,2} + q_{j,3})$$

subject to $\sum_{j,k} q_{j,k} = 1$, $q_{j,k} \geq 0$, and the eight observational equalities above.

5.1.3.5 Closed-form solution

The LP has a closed-form solution: each bound is a linear combination of the $p_{yx.z}$ whose gap from the objective is provably non-negative.

$$P(Y=1 \mid \mathrm{do}(X=0)) \;\geq\; \max \begin{cases} p_{10.0} \\ p_{10.1} \\ p_{10.0} + p_{11.0} - p_{00.1} - p_{11.1} \\ p_{01.0} + p_{10.0} - p_{00.1} - p_{01.1} \end{cases}$$

$$P(Y=1 \mid \mathrm{do}(X=0)) \;\leq\; \min \begin{cases} 1 - p_{00.0} \\ 1 - p_{00.1} \\ p_{10.0} + p_{11.0} + p_{01.1} + p_{10.1} \\ p_{01.0} + p_{10.0} + p_{10.1} + p_{11.1} \end{cases}$$

The first two lower bounds and upper bounds are the natural bounds; the extra terms make the LP bounds tighter.

Why is $p_{10.0} + p_{11.0} - p_{00.1} - p_{11.1}$ a valid lower bound? Substituting,

$$p_{10.0} + p_{11.0} - p_{00.1} - p_{11.1} = q_{0,2} + q_{0,3} + q_{1,2} + q_{2,3} - q_{0,0} - q_{0,1} - q_{2,0} - q_{1,1}.$$

Subtracting from the objective,

$$\sum_j (q_{j,2} + q_{j,3}) - (p_{10.0} + p_{11.0} - p_{00.1} - p_{11.1}) = q_{2,2} + q_{3,2} + q_{1,3} + q_{3,3} + q_{0,0} + q_{0,1} + q_{2,0} + q_{1,1} \;\geq 0.$$

Every closed-form bound is certified the same way (this is LP duality: the coefficients are a feasible dual solution).

5.1.4 Back-door with partially observed covariates

Motivation. Bound an effect when the admissible set is only partly measured but the marginal of the missing part is known.

Suppose $\{W, U\}$ is back-door admissible ($W \to X$, $W \to Y$, $U \to X$, $U \to Y$, W–U related, $X \to Y$), W observed, U not:

$$P(y \mid \mathrm{do}(x)) = \sum_{w,u} P(y \mid x, w, u) P(w, u) = \sum_{w,u} \frac{P(x, y, w, u) P(w, u)}{P(x, w, u)}.$$

Given $P(X, Y, W)$ and $P(U)$ but not $P(X, Y, W, U)$, the naive (natural) bound is $P(x, y) \leq P(y \mid \mathrm{do}(x)) \leq 1 - P(x, y')$, where $y' = 1 - y$, $x' = 1 - x$ denote complements. Li and Pearl (2022) tighten it:

$$a(y; x) = \min_{a_{w,u}, b_{w,u}, c_{w,u}} \sum_{w,u} \frac{a_{w,u}\, b_{w,u}}{c_{w,u}}, \qquad b(y; x) = \max_{a_{w,u}, b_{w,u}, c_{w,u}} \sum_{w,u} \frac{a_{w,u}\, b_{w,u}}{c_{w,u}},$$

where $a_{w,u}$, $b_{w,u}$, $c_{w,u}$ stand for the unknown $P(x, y, w, u)$, $P(w, u)$, $P(x, w, u)$, subject to, for all $w \in \mathfrak{X}_W$ and $u \in \mathfrak{X}_U$,

$$\begin{gathered} \textstyle\sum_u a_{w,u} = P(x, y, w), \qquad \sum_u b_{w,u} = P(w), \qquad \sum_u c_{w,u} = P(x, w), \\ b_{w,u} \geq c_{w,u} \geq a_{w,u} \quad (\text{since } P(x, y, w, u) \leq P(x, w, u) \leq P(w, u)), \\ \max\{0, P(x, y, w) + P(u) - 1\} \leq a_{w,u} \leq \min\{P(x, y, w), P(u)\}, \\ \max\{0, P(w) + P(u) - 1\} \leq b_{w,u} \leq \min\{P(w), P(u)\}, \\ \max\{0, P(x, w) + P(u) - 1\} \leq c_{w,u} \leq \min\{P(x, w), P(u)\}. \end{gathered}$$

The last three lines are Fréchet bounds for a joint probability given its two marginals. (In fact $a_{w,u}$ alone is a sufficient parametrization.)

5.2 Soft interventions

Motivation. The central question is how to evaluate realistic policies, which change a variable's mechanism to a conditional or stochastic rule rather than fixing it to a constant, and how the do-calculus generalizes to such soft interventions.

5.2.1 Motivation

Motivation. Show why atomic $\text{do}(x)$ interventions are often not the policies we can implement.

A tutoring program. For each student we observe the previous GPA W, motivation Z (low/high), tutoring X, and the final GPA Y. Motivation depends on previous GPA (and unobserved factors); students get tutoring depending on motivation; the final GPA depends on previous GPA, motivation and tutoring: $W \to Z \to X \to Y$, $W \to Y$, $Z \to Y$.

With large data and ML, $P(y \mid w, z, x)$ can be predicted with small error. But this distribution reflects the current, *natural regime* (regime: the mode a system operates in). Decisions intervene on the regime, so we must predict GPA under a hypothetical, unrealised regime.

What we know so far. From $P(W, Z, X, Y)$ and $\mathcal{G}$ we can identify

$$\begin{aligned} P(y \mid \text{do}(x)) &= \sum_z P(y \mid \text{do}(x), z) P(z \mid \text{do}(x)) && \text{condition on } Z \\ &= \sum_z P(y \mid \text{do}(x), z) P(z) && \text{Rule 3: } (Z \perp_d X)_{\mathcal{G}_{\overline{X}}} \\ &= \sum_z P(y \mid x, z) P(z) && \text{Rule 2: } (Y \perp_d X \mid Z)_{\mathcal{G}_{\underline{X}}}. \end{aligned}$$

Suppose $P(Y = 1 \mid \text{do}(X = 1)) > P(Y = 1)$: we predict improvement if $\text{do}(X = 1)$ is implemented. But what does $\text{do}(X = 1)$ mean in the real world? Mandatory tutoring for every single student. Is this realizable?

Implementing do-interventions. Even when the effect is identifiable, resources may be insufficient (not enough teachers and hours), and effectiveness cannot be guaranteed (non-compliance). For practicality we care about realizable interventions:

Do-like intervention	Realistic intervention
make sure no one smokes	reduce tobacco consumption to 20% of current
provide treatment to all patients	treat iff a patient is in critical condition
move a robotic arm to coordinate (x, y, z)	move it to (x, y, z) with a small error
mark all applicants as males	mark all applicants as males on paper

5.2.2 Canonical types of regimes

Motivation. Classify interventions by the new mechanism they install.

- **Idle/natural:** $X = f_X(\mathbf{pa}_X, u_X)$ (no intervention).
- **Hard/atomic:** $X = x$, $\text{do}(X = x)$, the type considered so far. Every student gets tutoring.
- **Conditional:** $X = g(\mathbf{w})$ sets X to the output of a function of observable variables $\mathbf{W}$. Students get tutoring iff they have a low GPA.
- **Stochastic:** $X \sim P^*(x \mid \mathbf{w})$ makes X follow a given distribution conditional on $\mathbf{W}$. Students with low GPA enter a raffle for 80% of the spots, other interested students for the remaining 20%.

We write σ_X for such an intervention and $P(\cdot;\sigma_X)$ for the induced distribution.

Changes in regime. The unperturbed world, where data are generated, is the natural regime with graph $\mathcal{G}$. An intervention creates a new regime differing in the mechanisms of the intervened variables. Its graph $\mathcal{G}_{\sigma_X}$ is drawn with a regime indicator (a small square σ_X) pointing into the intervened variables, and edges into X replaced according to the new mechanism.

5.2.3 Reduction to known machinery

Motivation. See when soft-intervention effects follow from hard-intervention results.

Can identification of soft (conditional, stochastic) interventions be reduced to that of hard interventions? **A reducible example.** Suppose $P(X=1 \mid Z=1)=0.3$: only 30% of motivated students get tutoring. What if this were doubled, σ_X: $P^*(X=1 \mid Z=1)=0.6$? In $\mathcal{G}_{\sigma_X}$, X's parents remain $\{Z\}$ but its mechanism changes. Then

$$\begin{aligned} P(y;\sigma_X) &= \sum_z P(y \mid z;\sigma_X)P(z;\sigma_X) = \sum_z P(y \mid z;\sigma_X)P(z) \\ &= \sum_{z,x} P(y \mid z,x;\sigma_X)P(x \mid z;\sigma_X)P(z) = \sum_{z,x} P(y \mid z,x;\sigma_X)P^*(x \mid z)P(z) \\ &= \sum_{z,x} P(y \mid z,x)P^*(x \mid z)P(z). \end{aligned}$$

Z is not affected by changing X's mechanism; the new assignment is $P^*(x \mid z)$; and given Z and X, Y responds as in the natural regime. Is a do-calculus-like derivation always sufficient?

An intentionally wrong derivation. A patient receives treatment X_1; a second physician observes $Z=z$ and decides a second treatment X_2; finally the patient survives or not (Y). Graph: $X_1 \to Z$, $X_1 \to X_2$, $X_1 \to Y$, $Z \to X_2$, $X_2 \to Y$, $X_1 \leftrightarrow Z$, $Z \leftrightarrow Y$. Consider fixing $X_1=x_1$ and $X_2=g(x_1,z)$, i.e. $\mathrm{do}(x_1), \mathrm{do}(X_2=g(x_1,z))$, or $\sigma_{\mathbf{X}}=\{X_1=x_1, X_2=g(x_1,z)\}$. One might write

$$P(y \mid \mathrm{do}(x_1), \mathrm{do}(X_2=g(x_1,z))) = P(y \mid x_1, \mathrm{do}(X_2=g(x_1,z))),$$

citing Rule 2 with $(Y \perp_d X_1 \mid X_2)$ in $\mathcal{G}_{\overline{X_2}\underline{X_1}}$. This is wrong: under $\mathrm{do}(X_2=g(x_1,z))$ the edges into X_2 remain active, so $\mathcal{G}_{\overline{X_2}\underline{X_1}}$ is not the right graph. The check must use $\mathcal{G}_{\sigma_{X_2}\underline{X_1}}$, where separation fails because, given the intervention on X_2, the path $X_1 \leftrightarrow Z \to X_2 \to Y$ is open.

Where do-calculus intuition breaks. Under do-interventions the intervened variables lose all dependence on their parents; soft interventions may keep some or all dependencies on parents, change their distribution, or even add new dependencies.

5.2.4 σ-calculus

Motivation. State inference rules for soft interventions that consult both pre- and post-interventional graphs.

Theorem 5.5 (σ-calculus, simplified; Correa and Bareinboim 2020)

For disjoint $\mathbf{X},\mathbf{Y},\mathbf{Z} \subseteq \mathbf{V}$ and disjoint $\mathbf{T},\mathbf{W} \subseteq \mathbf{V} \setminus (\mathbf{Z} \cup \mathbf{Y})$:

Rule 1 (insertion/deletion of observations) $P(\mathbf{y} \mid \mathbf{w},\mathbf{t};\sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w};\sigma_{\mathbf{X}})$ if $(\mathbf{Y} \perp_d \mathbf{T} \mid \mathbf{W})$ in $\mathcal{G}_{\sigma_{\mathbf{X}}}$.

Rule 2 (change of regimes under observation) $P(\mathbf{y} \mid \mathbf{w},\mathbf{x};\sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w},\mathbf{x})$ if $(\mathbf{Y} \perp_d \mathbf{X} \mid \mathbf{W})$ in $\mathcal{G}_{\sigma_{\mathbf{X}}\underline{\mathbf{X}}}$ and in $\mathcal{G}_{\underline{\mathbf{X}}}$.

Rule 3 (change of regimes without observation) $P(\mathbf{y} \mid \mathbf{w};\sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w})$ if $(\mathbf{Y} \perp_d$

$\mathbf{X} \mid \mathbf{W})$ in $\mathcal{G}_{\sigma_{\mathbf{X}}\overline{\mathbf{X}(\mathbf{W})}}$ and in $\mathcal{G}_{\overline{\mathbf{X}(\mathbf{W})}}$.

Proof [Supplement]. The proof parallels that of do-calculus (Chapter 3), with the regime indicator now selecting between the natural mechanism and the stochastic policy $\sigma_{\mathbf{X}}$. **Rule 1** is d-separation in $\mathcal{G}_{\sigma_{\mathbf{X}}}$, to which the post-regime distribution is Markov. For **Rules 2 and 3**, add a regime switch F_X with $F_X \to X$; the two sides correspond to $F_X = \sigma$ and $F_X =$ idle. Exactly as in the proof of Rules 2 and 3 of do-calculus, $F_X \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{W}, \mathbf{X}$ (Rule 2) or $F_X \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{W}$ (Rule 3) holds if the corresponding separation holds with the outgoing (respectively incoming) edges of $\mathbf{X}$ removed. Because a soft regime may add new parents of $\mathbf{X}$, the separation must hold in the graph of each regime, which is why both $\mathcal{G}_{\sigma_{\mathbf{X}}\ldots}$ and $\mathcal{G}_{\ldots}$ appear in the conditions. ■

Full version. With an outer regime $\sigma_{\mathbf{X}}$ and a regime change on $\mathbf{Z}$ from $\sigma_{\mathbf{Z}}$ to $\sigma'_{\mathbf{Z}}$:

- Rule 1: $P(\mathbf{y} \mid \mathbf{w}, \mathbf{t}; \sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w}; \sigma_{\mathbf{X}})$ if $(\mathbf{Y} \perp_d \mathbf{T} \mid \mathbf{W})$ in $\mathcal{G}_{\sigma_{\mathbf{X}}}$;
- Rule 2: $P(\mathbf{y} \mid \mathbf{w}, \mathbf{z}; \sigma_{\mathbf{Z}}, \sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w}, \mathbf{z}; \sigma'_{\mathbf{Z}}, \sigma_{\mathbf{X}})$ if $(\mathbf{Y} \perp_d \mathbf{Z} \mid \mathbf{W})$ in $\mathcal{G}_{\sigma_{\mathbf{X}}\sigma_{\mathbf{Z}}\underline{\mathbf{Z}}}$ and $\mathcal{G}_{\sigma_{\mathbf{X}}\sigma'_{\mathbf{Z}}\underline{\mathbf{Z}}}$;
- Rule 3: $P(\mathbf{y} \mid \mathbf{w}; \sigma_{\mathbf{Z}}, \sigma_{\mathbf{X}}) = P(\mathbf{y} \mid \mathbf{w}; \sigma'_{\mathbf{Z}}, \sigma_{\mathbf{X}})$ if $(\mathbf{Y} \perp_d \mathbf{Z} \mid \mathbf{W})$ in $\mathcal{G}_{\sigma_{\mathbf{X}}\sigma_{\mathbf{Z}}\overline{\mathbf{Z}(\mathbf{W})}}$ and $\mathcal{G}_{\sigma_{\mathbf{X}}\sigma'_{\mathbf{Z}}\overline{\mathbf{Z}(\mathbf{W})}}$.

5.2.4.1 Comparison with do-calculus

Remark. Under $\mathrm{do}(x)$, X is deterministic, so $P(y \mid \mathrm{do}(x), w) = P(y \mid \mathrm{do}(x), w, x)$. Under a non-atomic intervention this is not necessarily so: for $X \to Y$, $P(y; \sigma_X) \neq P(y \mid x; \sigma_X)$ in general.

	do-calculus	σ-calculus
Rule 1	$P(y \mid \mathrm{do}(x), w, t) = P(y \mid \mathrm{do}(x), w)$ if $(Y \perp_d T \mid W, X)_{\mathcal{G}_{\overline{X}}}$	$P(y \mid w, t; \sigma_X) = P(y \mid w; \sigma_X)$ if $(Y \perp_d T \mid W)_{\mathcal{G}_{\sigma_X}}$
Rule 2	$P(y \mid \mathrm{do}(x), w) = P(y \mid x, w)$ if $(Y \perp_d X \mid W)_{\mathcal{G}_{\underline{X}}}$	$P(y \mid x, w; \sigma_X) = P(y \mid x, w)$ if $(Y \perp_d X \mid W)$ in $\mathcal{G}_{\sigma_X \underline{X}}$ and $\mathcal{G}_{\underline{X}}$
Rule 3	$P(y \mid \mathrm{do}(x), w) = P(y \mid w)$ if $(Y \perp_d X \mid W)_{\mathcal{G}_{\overline{X(W)}}}$	$P(y \mid w; \sigma_X) = P(y \mid w)$ if $(Y \perp_d X \mid W)$ in $\mathcal{G}_{\sigma_X \overline{X(W)}}$ and $\mathcal{G}_{\overline{X(W)}}$

In Rule 1 the relevant graph depends on the specification of the intervention; in Rules 2 and 3 the independence must hold in both the pre- and post-interventional graphs.

Revisiting Rule 2. For the sequential-treatment example,

$$\begin{aligned} P(y \mid \mathrm{do}(x_1), \mathrm{do}(X_2 = g(x_1, z))) &= P(y \mid x_1, \mathrm{do}(x_1), \mathrm{do}(X_2 = g(x_1, z))) \\ &= P(y \mid x_1; \sigma_{X_1, X_2}) \overset{?}{=} P(y \mid x_1; \sigma_{X_2}). \end{aligned}$$

To invoke Rule 2 we must check the absence of a back-door path from X_1 to Y in two graphs: the one where X_2 is set by g ignoring Z, and the one where X_2 still listens to X_1 and Z. It holds only in the first, so the rule is not applicable.

5.2.4.2 Another intervention

Resources are limited, so tutoring is focused on those who need it most: students with low GPA must get tutoring, and only they can, i.e. $P^*(X = 1 \mid W = 0) = 1$, $P^*(X = 1 \mid W = 1) = 0$, a conditional intervention ($W = 0$ denoting low GPA). In $\mathcal{G}_{\sigma_X}$, X now depends on W (not on Z):

$$\begin{aligned} P(y; \sigma_X) &= \sum_{w,z,x} P(y \mid x, w, z; \sigma_X) P(x \mid w, z; \sigma_X) P(w, z; \sigma_X) \\ &= \sum_{w,z,x} P(y \mid x, w, z; \sigma_X) P(x \mid w; \sigma_X) P(w, z; \sigma_X) \qquad \text{Rule 1} \end{aligned}$$

$$= \sum_{w,z,x} P(y \mid x, w, z) P(x \mid w; \sigma_X) P(w, z; \sigma_X) \qquad \text{Rule 2}$$

$$= \sum_{w,z,x} P(y \mid x, w, z) P^*(x \mid w) P(w, z) \qquad \text{Rule 3.}$$

5.2.5 C-factors and soft interventions

Motivation. Identify soft-intervention effects systematically with c-factors.

For $\mathcal{G}$: $Z \to X$, $Z \to Y$, $X \to Y$,

$$P(z, x, y) = \sum_{u_Z} P(z \mid u_Z) P(u_Z) \sum_{u_X} P(x \mid z, u_X) P(u_X) \sum_{u_Y} P(y \mid x, z, u_Y) P(u_Y) = Q[Z]\, Q[X]\, Q[Y].$$

Under $\mathcal{M}' = \mathcal{M}_{\sigma_X}$ with diagram $\mathcal{G}' = \mathcal{G}_{\sigma_X}$, only X's mechanism changes:

$$P(z, x, y; \sigma_X) = \sum_{u_Z} P(z \mid u_Z) P(u_Z) \sum_{u'_X} P'(x \mid z, u'_X) P(u'_X) \sum_{u_Y} P(y \mid x, z, u_Y) P(u_Y)$$

$$= Q[Z]\, Q[X; \sigma_X]\, Q[Y].$$

In general, $P(\mathbf{y}; \sigma_{\mathbf{X}}) = \sum_{\mathbf{d} \setminus \mathbf{y}} \prod_{\mathbf{D}_i \in \mathcal{C}(\mathbf{D})} Q[\mathbf{D}_i]$, the product over the c-components of $\mathcal{G}_{\sigma_{\mathbf{X}}}$ restricted to the relevant ancestors; $Q[\mathbf{X}; \sigma_{\mathbf{X}}]$ is given by the definition of $\sigma_{\mathbf{X}}$, and every other factor must be identified from $P(\mathbf{V})$, e.g. with IDENTIFY (Chapter 3).

Example 5.2 (Systematic identification 1)

$\mathcal{G}$: $W \to Z$, $X \to Z \to Y$, $X \leftrightarrow Y$; intervention $\sigma_X = P^*(X \mid W)$, so $\mathcal{G}_{\sigma_X}$ has $W \to X$ and loses $X \leftrightarrow Y$. C-components: in $\mathcal{G}$, $\mathbf{C}_1 = \{X, Y\}$, $\mathbf{C}_2 = \{W\}$, $\mathbf{C}_3 = \{Z\}$; in $\mathcal{G}_{\sigma_X}$, $\mathbf{D}_1 = \{Y\}$, $\mathbf{D}_2 = \{W\}$, $\mathbf{D}_3 = \{Z\}$, $\mathbf{D}_4 = \{X\}$. Subgoal: identify each $Q[\mathbf{D}_i]$; $Q[\mathbf{D}_4] = Q[X; \sigma_X]$ is given by the intervention. IDENTIFY$(\{Y\}, \{X, Y\}, Q[T], \mathcal{G})$: $A = An(\{Y\})_{\mathcal{G}[\{X,Y\}]} = \{Y\} = C$, so $Q[Y] = \sum_x Q[X, Y]$. IDENTIFY$(\{W\}, \{W\}, \cdot)$ and IDENTIFY$(\{Z\}, \{Z\}, \cdot)$ return immediately. Hence

$$P(y; \sigma_X) = \sum_{w,x,z} Q[X; \sigma_X]\, Q[Z]\, Q[W]\, Q[Y] = \sum_{w,x,z} Q[X; \sigma_X]\, Q[Z]\, Q[W] \sum_{x'} Q[X, Y]$$

$$= \sum_{w,x,z} P^*(x \mid w)\, P(z \mid x, w)\, P(w) \sum_{x'} P(x')\, P(y \mid x', z, w).$$

Example 5.3 (Systematic identification 2: failure)

$\mathcal{G}$: $W \to Z \to X \to Y$, $W \leftrightarrow X$, $W \leftrightarrow Y$; intervention $\sigma_X = P^*(X \mid Z)$. C-components: in $\mathcal{G}$, $\mathbf{C}_1 = \{W, X, Y\}$, $\mathbf{C}_2 = \{Z\}$; in $\mathcal{G}_{\sigma_X}$, $\mathbf{D}_1 = \{Y, W\}$, $\mathbf{D}_2 = \{X\}$, $\mathbf{D}_3 = \{Z\}$. IDENTIFY$(\{W, Y\}, \{W, X, Y\}, Q[T], \mathcal{G})$: $A = An(\{W, Y\})_{\mathcal{G}[\{W,X,Y\}]} = \{W, X, Y\} \neq C$ and $A = T$, so FAIL. (Recall $Q[\mathbf{C}_j] = \prod_{V_i \in \mathbf{C}_j} Q[\mathbf{V}^{\leq i}]/Q[\mathbf{V}^{\leq i-1}]$ and $Q[\mathbf{V}^{\leq i}] = \sum_{\mathbf{v}^{>i}} Q[\mathbf{V}]$.)

5.2.6 Summary

- In decision-making settings, where interventions must be implemented, non-atomic (conditional and stochastic) interventions are needed.
- In the spirit of do-calculus, analogous rules consulting both pre- and post-interventional graphs identify the effects of soft interventions.
- Soft-intervention effects can also be identified by systematic procedures based on c-factors.